

OptiCrop: Project Executable Files & Run Instructions

1. Repository

GitHub: <https://github.com/sriramakanthpendikatla/OptiCrop>

2. Executable Files in this Project

- `app.py` — Flask web application; run this file to start the server.
- `notebook/OptiCrop_Model.ipynb` — full pipeline: EDA, preprocessing, model training, evaluation, K-Means clustering, model saving, conclusion.
- `data_preprocessing/preprocessing.ipynb` — preprocessing and model-building notebook (extracted from the full pipeline).
- `visualization_analisys/visualization.ipynb` — exploratory data analysis and visualization notebook.
- `model.pkl`, `scaler.pkl`, `label_encoder.pkl` — serialized trained artifacts loaded by `app.py` at runtime.

3. Setup Instructions

Clone the repository:

```
git clone https://github.com/sriramakanthpendikatla/OptiCrop.git
cd OptiCrop
```

Create and activate a virtual environment:

```
python -m venv venv
source venv/bin/activate      # On Windows: venv\Scripts\activate
```

Install dependencies:

```
pip install -r requirements.txt
```

4. Running the Web Application

```
python app.py
```

Then open **<http://127.0.0.1:5000>** in a browser. Enter soil nutrient values (N, P, K), temperature, humidity, pH, and rainfall, and submit the form to receive a recommended crop.

5. Running the Notebooks

```
jupyter notebook
```

Open `notebook/OptiCrop_Model.ipynb` for the complete end-to-end pipeline, or the split notebooks (`data_preprocessing/preprocessing.ipynb` and `visualization_analisys/visualization.ipynb`) for the corresponding sections.

6. Model Performance Summary

- Accuracy: 97.27%
- Precision: 97.42%
- Recall: 97.27%
- F1-Score: 97.25%
- ROC-AUC: 0.9998